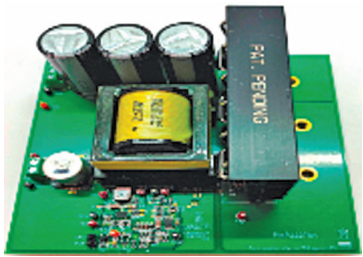


Interesting Reference Designs

Title of Reference Design	Reference Design for a SEPIC Power Supply	Reference Design for a 200W SEPIC Power Supply
Image		
Key Attractions	Reference design for a SEPIC converter featuring a universal PWM converter. The reference design offers integrated control, output adjustment, monitoring, and multiple protection features. The reference design offers resistor-selectable switching frequency from 100kHz to 1.4MHz.	Reference design for a non-isolated adjustable 5V to 200V power supply with a maximum output current of 4A. The reference design works with 40V input and offers an efficiency of up to 93.7%.
Highlights	The ISL8130EVAL3Z from Renesas is a reference design for a DC-DC converter which can deliver up to 10W of power. The reference design features a SEPIC topology and employs universal PWM controller ISL8130. This reference design has a small size measuring just 4.5cm x 3.15cm. The power supply can take 5.5 to 16V input and provide a constant output of 10V. The ISL8130EVAL3Z reference design also offers a resistor-selectable switching frequency from 100kHz to 1.4MHz, which makes it user adjustable and also provides a simple voltage mode control with fast transient response. The power supply also features a power good signal that indicates the user that the supply has stabilised its output voltages and passed all its internal self-tests, making it suitable for use in sensitive applications, such as power supply for medical devices. The reference design offers multiple safety features, such as overcurrent protection, overvoltage protection, and undervoltage protection.	PMP22270 is a reference design for a 5V to 200V power supply from TI. It can provide an output of up to 4A. The output voltage is set using a 0% to 100% duty-cycle, pulse-width-modulation (PWM) input. The reference design can work with input of 38V to 42V and provides a maximum output of 200V with an efficiency of up to 93.7%. The design enables low-input-ripple voltage and low EMI compared to flyback converters. This reference design is based on LM5022-Q1, which is a high-voltage, low-side N-channel MOSFET controller ideal for use in boost and SEPIC regulators. It contains all the features needed to implement single-ended primary topologies. The reference design provides 3x to 5x higher output power than traditional SEPICs and enables low-input-ripple voltage and low EMI compared to flyback converters. Output voltage regulation is based on current-mode control, which eases the design of loop compensation while providing inherent input voltage feed-forward. The reference design is suitable for surgical kits, toys, and household equipment.
Application	The reference design is suitable for use as a power supply for microprocessors, ASICs, Ethernet routers, switches, medical and industrial instruments, etc.	The reference design is suitable for use as a power supply for surgery systems, industrial instruments, toys, etc.
OEM Brand	Renesas	TI
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-10w-power-supply-working-on-sepic-topology	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-200w-adjustable-power-supply